

Akshit Bansal

SOFTWARE DEVELOPMENT ENGINEER II

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Work Experience

CRED – UCMS Platform & Promotions (Ads & Audiences)

Bangalore, India

SOFTWARE DEVELOPMENT ENGINEER II

Mar. 2024 - Present

- Built **'DataLens,'** a centralized Model Context Protocol (MCP) server that standardizes data retrieval from Mixpanel; adopted across the org as the core data fetcher for multiple autonomous agents; other engineers has extended it to pull data from other data sources as well.
- Developed an autonomous **'Inference-Guard'** GitHub Action utilizing agentic coding to monitor critical config changes; auto-generates PRs to synchronize downstream data layers, preventing silent schema drift.
- Architected a natural language **'Insights Engine'** within the UCMS platform; integrated LLM agents via MCP to allow non-technical users to query campaign performance metrics via chat, bypassing complex UI dashboards.
- Built **Campaign monitoring** module end-to-end: YAML-configurable metrics module ingesting from Victoria Metrics, Mixpanel, and Databricks, plus a BE+FE dashboard with per-campaign and default Slack alerts on top of metrics.
- Led **"store" LOB templates onboarding** to unified campaign management service, enabling 2,580 campaigns/month at 80% coverage; grew platform-wide coverage from 25% to 57% in 6 months and saved 250+ man-hours/month of campaign-ops effort.
- Architected an in-house **Geolocation service** with lat/long-based city resolution at **p99 sub-7ms under 1.9k RPS peaks**; automated Maxmind S3 sync for eventually-consistent global location data.
- Optimized Adserver scalability via a Flink windowed-aggregation pipeline, **cutting database load by 1,000 TPS**; restored p99 latency from 30ms to 15ms via parallel reads and budget-data caching.
- Eliminated weekly OOM restarts by diagnosing and refactoring a critical Lucene commit leak, stabilizing service uptime.

CRED – Promotions Platform (Ads & Audiences)

Bangalore, India

SOFTWARE DEVELOPMENT ENGINEER I

Sep. 2022 - Mar. 2024

- Solely architected and shipped an in-house **Adserver** replacing a 3rd-party vendor — **saving \$300K+ (INR 2.8 Cr) annually** and dropping p99 latency **from 36ms to 12ms**.
- Acted as Architectural Lead for a new central **Reporting Service**, mentoring a junior engineer from design to deployment; successfully onboarded 162 merchants and saved ~2 hours of manual daily reporting effort.
- Built a high-throughput **audience-sync pipeline** (40GB/run) for major merchants; reduced Swiggy reactivation leakage **from 22% to 0.4%** and eliminated 2M+ wasted monthly syncs.
- Powered **RMG decile cuts** on top of the audience-sync layer, driving **~80% conversions on ~35% of original impressions** for targeted campaigns.
- Added **campaign-sync capabilities**: multi-rate support (CPR, CPT, CPA), hybrid frequency caps, and Zerkel targeting, driving **5%+ CTR and 3%+ ITR uplift** across the carousel.
- RCA'd Mordor JVM memory spikes to stable ~60% memory; split configs into prod/stage to streamline SSM management.

Udaan – Pharma Tech Supply Chain

Bangalore, India

FULL STACK ENGINEER

Jan. 2021 - Sep. 2022

- Engineered warehouse-automation modules including a weight-capturing system (**INR 7L/warehouse annual GMV increase**) and data-driven put-away algorithms that improved quality-check times by 10%.
- Shipped an in-house digital signature flow, reducing invoice re-upload pendency **from 80% to 0%** and eliminating physical paper costs.

Technical Projects

SELF-CORRECTING DIAGNOSTIC RAG PIPELINE

- Architected a stateful multi-agent RAG pipeline (**LangGraph, local Llama 3**) featuring an autonomous 'Critic Node' that programmatically catches hallucinations and forces cyclical revisions prior to output.
- Engineered a hybrid-search retrieval system (**Qdrant, FastEmbed**) for deterministic data extraction, and established LLMops evaluation pipelines (**Ragas**) to benchmark Context Precision and Faithfulness.

HARDWARE PROJECTS

- Designed and assembled a custom multi-channel physical audio mixer (popularly known as **Deej** in open source community) using Arduino Nano and rotary potentiometers to manage application-level audio via serial communication.

Skills

AI & Orchestration	LangGraph, Qdrant (Vector DB), Ragas (LLMOps), Agentic Workflows, MCP, Local Inference (Ollama)
Backend & Systems	Java, Kotlin, Python, Microservices, Distributed Systems, Lucene, Flink, Aerospike, Low-Latency Architectures
Data & Cloud	AWS (S3, Lambda), Databricks, Airflow, Mixpanel, SQL, Grafana, Datadog, Prometheus
Core Domains	Ad Tech, Campaign Management, Performance Optimization, Warehouse Automation, System Design Ownership

Education

Indian Institute of Information Technology Allahabad

Prayagraj, India

B.TECH IN INFORMATION TECHNOLOGY

2017 - 2021

- CGPA: 8.89 / 10.0**